

Deepak Pathak

Principal Engineer with 15+ years of experience designing and building scalable distributed systems, microservices architectures, and enterprise integration platforms across payments, logistics, and retail domains. Proven track record of leading architecture, driving cloud transformations, and building AI-enabled systems using modern LLM frameworks and APIs.

PROFESSIONAL EXPERIENCE

WiseTech Global/Blume Global — Principal /Lead Software Engineer

2022 – Present | USA

- Architected and developed microservices supporting logistics workflows, processing 10K+ shipment events/day per client across multiple enterprise integrations.
- Built Generative AI solutions leveraging OpenAI/Claude GPT models, LangChain, and Retrieval-Augmented Generation (RAG) architectures.
- Developed intelligent AI chatbots and virtual assistants for customer support and internal automation workflows.
- Implemented vector search solutions using Pgvector for semantic search and document retrieval systems.
- Created RESTful APIs and microservices for AI model orchestration and enterprise integrations.
- Designed prompt engineering strategies to improve AI response quality, contextual accuracy, and workflow automation.
- Designed resilient backend systems achieving 99.9%+ system uptime.
- Collaborated with cross-functional teams to deliver scalable architecture supporting rapid feature expansion.

FIS, USA — Lead Software Engineer/ Software Engineer Specialist

2016 – 2022 | USA

- Led architecture and development of inventory systems handling high transaction volumes (100K+ events/day)
- Designed and executed cloud migration strategy (DB2 → MySQL, on-prem → AWS), reducing infrastructure costs by ~20%
- Introduced CI/CD pipelines and automation, reducing deployment time from hours to minutes
- Improved system performance by optimizing APIs and database queries, reducing latency by ~30%
- Developed monitoring and alerting improvements reducing production incidents by ~25%
- Mentored 10+ engineers and led cross-functional teams across onsite/offshore delivery models

FIS, India — Development Supervisor/Project Lead

2009 -2016 | India

- Led teams of 8–12 engineers delivering enterprise financial systems with high reliability and scalability
- Delivered multiple large-scale releases with consistent on-time delivery and improved code quality standards
- Reduced defect leakage by ~20% through improved testing and review practices
- Owned end-to-end delivery including architecture, development, and stakeholder communication

Pleasanton, CA

(510) 585-6271

deepak.pathakemail@gmail.com

LinkedIn: [deepak pathak](#)

GitHub: [dpathak-ai](#)

SKILLS

Languages & Frameworks - Java, Python, Spring Boot, Spring JPA, Hibernate, REST APIs, Fast API, React, Node.js

Cloud & DevOps - AWS (EC2, S3, RDS, Lambda), Docker, Kubernetes, GCP (VM, GKE) CI/CD (Jenkins, GitHub Actions), Distributed Caching (Redis), Observability Grafana, Prometheus.

AI & Data – LLM APIs, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, OpenAI SDK, CrewAI, LangGraph, MCP Server, Google ADK

Messaging & Integration - Kafka, REST/SOAP Services, Event-Driven Architecture

Database - MySQL, DB2, SQL, Stored Procedures

Architecture - Microservices, Distributed Systems, System Design, API Design

CERTIFICATIONS

Certificate Program in Data Science 2021

University of Berkeley Extension, CA, USA

Project Management Professional (PMP) 2016

Project Management Institute, US

EDUCATION

Master's In Computer Application

2002-2005, HNB Garhwal Central University, Sri Nagar, UK, India

Bachelor's In Computer Application

1998-2001, CCS University Meerut, UP, India

KEY PROJECTS

AI-Powered Document Processing System

- Designed a system using LLM APIs and vector databases to process and query integration documents
- Implemented RAG-based architecture improving information retrieval accuracy by ~35%
- Reduced manual document processing effort by ~30% through automation

Logistics Microservices & Integration System

- Led integration of core Blume entities with WiseTech Transport Booking module, enabling seamless data flow and operational efficiency.
- Enhanced Logistics APIs supporting high-throughput integrations, improving scalability and reducing response latency by ~20%
- Designed and implemented WiseTech E-Request integration within the Blume–WiseTech ecosystem, streamlining support workflows and eliminating dependency on Salesforce CRM.
- Implemented secure Single Sign-On (SSO) using OAuth standards across WiseTech and Blume applications, improving user experience and strengthening authentication mechanisms.

Cloud Migration & Modernization

- Defined and executed cloud migration strategy for catalog and order processing systems, transitioning from DB2 to MySQL and from on-premise data centers to Amazon Web Services (AWS).
- Evaluated and implemented migration approaches including lift-and-shift and data replication using AWS Database Migration Service and Attunity, ensuring minimal downtime and data integrity.
- Collaborated with DevOps and CloudOps teams to provision AWS infrastructure and establish robust CI/CD pipelines for automated deployments.
- Resolved complex migration challenges across infrastructure, networking, application layers, and database components (SQL objects and queries), ensuring seamless integration with downstream systems.

Inventory Replenishment Microservices

- Designed and developed two scalable microservices and a frontend application to support supply chain operations for Lawson Japan, ensuring high availability and operational continuity.
- Built microservices using Spring Boot with asynchronous file processing via Spring Async, improving throughput and reducing processing latency.
- Developed responsive frontend applications using Vue.js with multilingual support, enhancing usability for diverse user groups.
- Leveraged technologies including Apache iBATIS, Redis Cache, and Amazon Web Services (AWS) to optimize performance, caching, and scalability across distributed systems.

Catalog and Order Processing System

- Developed and scaled a Catalog and Order Processing System supporting digital and brick-and-mortar channels, processing ~100K orders/day and managing 10K+ product SKUs, enabling end-to-end workflows from master data ingestion to automated catalog and order generation
- Architected web applications, REST APIs, and batch processing (scheduled/on-demand), improving system throughput by ~20% and reducing processing latency by ~25%.
- Optimized application logic and SQL queries, conducted rigorous code reviews, and implemented end-to-end testing, reducing production defects by ~25% and improving system reliability (achieving ~99.9% uptime).
- Modernized build and deployment pipelines, upgrading repositories and tooling, reducing release cycle time by ~40–60% and minimizing deployment-related incidents by ~20%.

ADDITIONAL INFORMATION

- Strong expertise in system design, distributed architectures, and enterprise integrations
- Experience leading global teams and driving technical strategy
- Passionate about building AI-enabled scalable systems and modern backend platforms